

PUBH 526

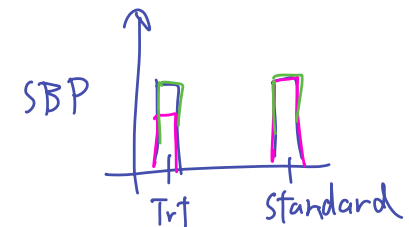
Topic 1, Part 2:
Introduction and Review

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Let's Design a Study...

You want to test a new drug for its efficacy in reducing systolic blood pressure

- How would you design such a study? ★ (RCT) Trt v.s. Control
★ Measure SBP after Trt.
- What would be your hypothesis? Trt lowering SBP compared to control
- What would data consistent with your hypothesis look like? ② Same Effect ① Trt lowering SBP
- What would data inconsistent with your hypothesis look like? Trt increasing SBP (not favorable) ✓ (favorable Effect)
- What kind of statistical test would you use?
★ ① two-sample t-test (comparing mean SBP between groups)
② ANOVA if >2 groups if adjusting for covariates



Design of Experiments

- Different disciplines have different applications, approaches, vocabulary, methods, history, literature
- Origins in agriculture
- Industry and engineering
- Health sciences
- Social sciences, including economics



Fundamental Principles

Randomization

Ensures fair comparisons by:

- ① distributing known/unknown factors across groups.
- ② ↑ validity of causal inference

- What is randomized? -- unit of randomization Ex) subjects, population, individuals, groups
 - Individual people
 - Classrooms at a school
 - Villages in a rural area
 - Counties in Indiana
- Measurement and analysis may occur at the unit of randomization or at higher or lower levels
 - Units of randomization, units of measurement/analysis, context Ex) clustering...
who/what *where outcomes are measured*
- Why randomize?
 - Balance out effects of “lurking” variables – **confounders, effect modifiers, mediators...**
 - Many of these **may be unknown**
 - Does “balance out” mean “eliminate”? *No. Goal: Make Trt v.s. Control Comparable on average*

Treatments

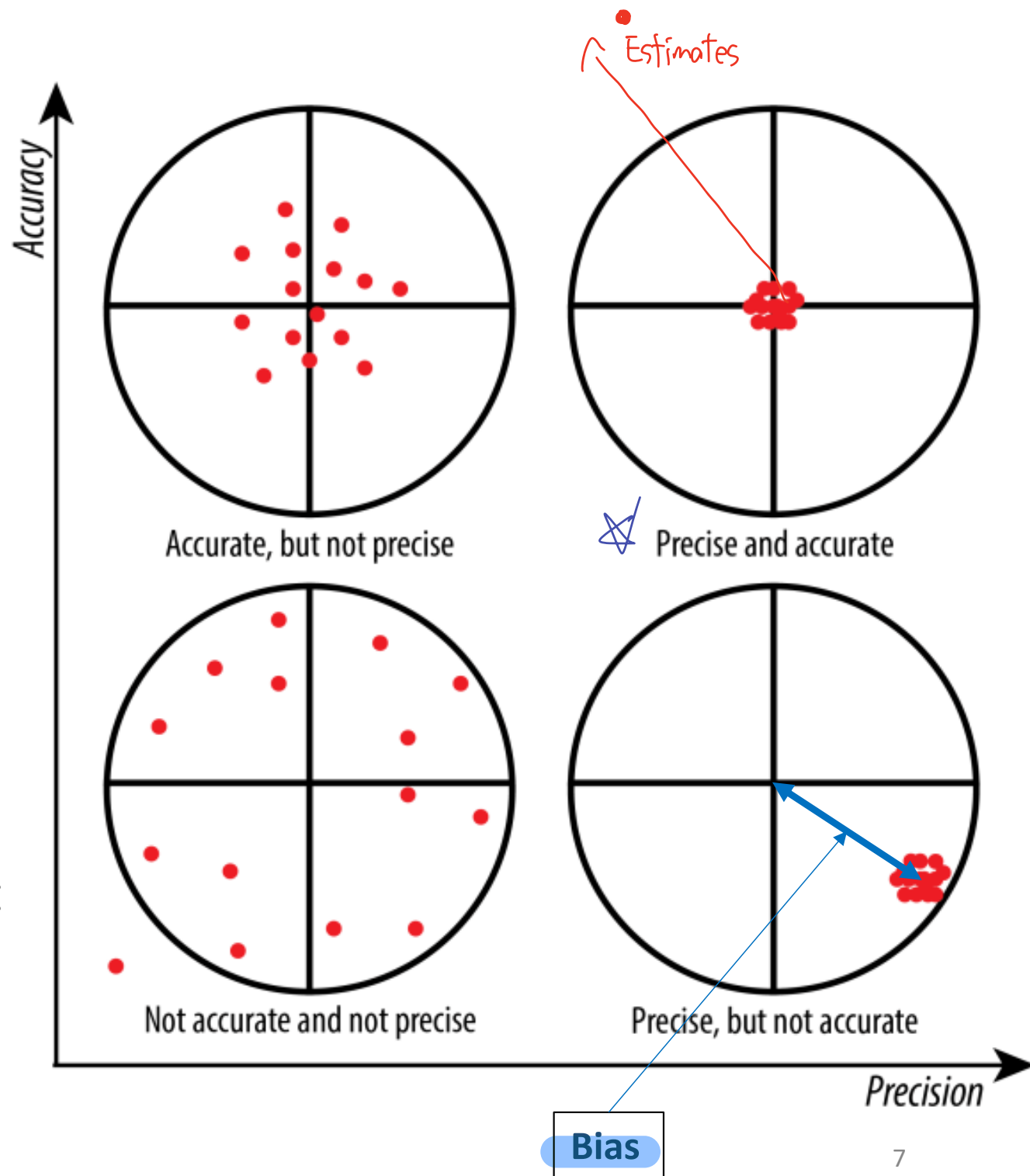
- In public health:
 - Treatment effects take time
 - The control treatment may not be obvious
- **Counterfactual:** what would have happened **if there were no intervention?**
 - Changes over time? – secular? seasonal? *Ex) Measuring depression? Seasonal in winter*
- Often difficult to separate the intended treatment from the delivery method
 - New drug vs. placebo
 - How about a health informational session?

Estimating Treatment Effects

- Accuracy is about hitting the target
- Precision is about spread
- What exactly is the target here? And what is an estimate?

Truth
Population Parameters → Targets

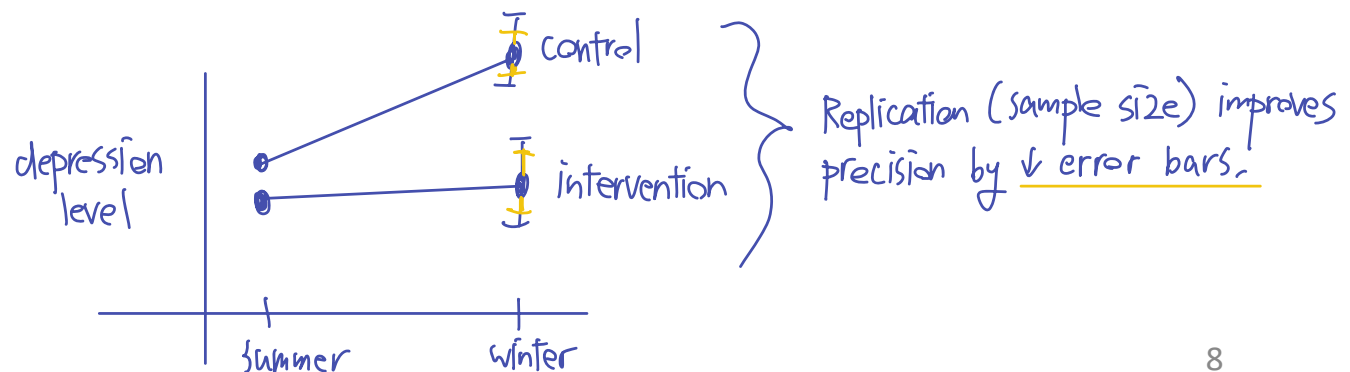
Estimate → Red Dots



◦ Replication $\uparrow$ Power

Replication

- Sample size $\uparrow$ probability of detecting a true effect if one exists ($\downarrow$ Type 2 error: false neg.)
- Source of statistical power
- A greater sample size improves the precision of your $\downarrow$ CI estimates, including your estimate of the error or background noise $\uparrow$ stability, $\downarrow$ sensitivity to random fluctuations
- Consider sample size at all levels of measurement, particularly at the level of randomization



Blocking ^{◦ Blocking ↑ precision}

- A way of dealing with “nuisance” factors
- Stratification
- Different blocks have different levels of the nuisance factor, but within a block, participants are more similar
- Examples?
 - **Agriculture:** Fields differ in soil fertility → block by field, randomize treatments within each field.
 - **Clinical trial:** Patients differ by sex or baseline health → block by sex/health category, randomize treatment within each block.
 - **Education:** Schools differ by size → block by school, randomize classrooms within each.

◦ Blinding ↓ bias

Blinding

- Why?

- Who?

Don't know if they
receive Trt or Control

Don't know which subjects
are in which group.

- Participants, researchers (many roles here), both (double blinding),

Either patient nor doctor knows

- How? (blinded, unblinded, codes only)

Everyone knows the Trt.
(when blinding is not possible)

Triple blinding
(+ analysts, reviewers)

- When?

↓ concealed

Ex) Placebo looks same as Trt pills

Blinding

- How do you randomize while maintaining **double blinding**?
 - **Roles:** PI, senior personnel, pharmacists, physicians, nurses, data collectors, specimen collectors, data managers/cleaners/analysts, participants, ... (**blind at different levels at different times**)
 - It must look and feel the same (identical) placebo pills/procedures
 - How to ensure consistency of treatments for the study duration?
 - Blinding must be broken if safety concern
 - How do you maintain participant safety?
 - Anticipate adverse outcomes, collect data equally in all study arms

- When is blinding not possible?

- Of participants? Of researchers?

when is blinding not possible?

- ↳ Giving foods to participants (they know what the food looks like) **Researchers/Staff**
- ↳ People preparing/sharing foods
- ↳ Community Interventions (Messages conveyed)
- ↳ Lifestyle Interventions Ex) diet, exercise... **Participants**

Internal and External Validity

- **Internal validity:** The degree to which the results are attributable to the **experimental treatment** and not some other explanation within study sample
- **External validity:** The extent to which the results of a study can be **generalized** outside the study

Feature	Internal Validity	External Validity
Concern	Is the causal relationship true <i>inside</i> the study?	Can results be applied <i>outside</i> the study?
Threatened by	Bias, confounding, measurement error	Lack of representativeness, artificial setting
Goal	Correct inference about cause & effect	Generalizability to real-world populations
Improved by	Randomization, blinding, replication	Diverse samples, pragmatic study design, replication across contexts

Internal Validity

IV comes first !
◦ If your study isn't valid inside, you can't generalize it! (EV)

- Internal validity allows us to make statements about causality
 - However, mechanisms may not be clear
- Randomization aims to balance “explanations” across treatment arms
 - At start of study and before
 - During the study (assuming “explanations” affect both groups similarly)
 - Implementation should be balanced as much as possible – and monitored
- (Double) blinding -- to minimize biases of researchers and participants
- Attrition and missing data
 - Differential across treatments?
 - If attrition is the similar across treatments, is everything ok?

External Validity

- How representative is this sample of the larger population?
 - Did participants elect to participate?
 - What was the burden of participation?
 - Context
- Is the larger population clearly defined?
- How will the larger / other populations react to the specific treatment / intervention?
- Will the treatment remain the same when scaled up? How about the quality of implementation?

Designs to Be Covered in This Course

- Completely randomized design
- Randomized complete block design
- Crossover design
- Cluster randomized design
- Factorial design
- Other topics...